

Task_1:-Data_Cleaning_and_Preparation

In Task 1, Firstly duplicate records were identified and removed to avoid repeated information in the dataset. Missing values were checked and handled appropriately to improve data quality. Column names were cleaned and standardized for easier analysis and coding. And, Some categorical values such as recommendation helpfulness and review reliability were converted into numeric format.

```

TASK 1:-DATA_CLEANING_AND_PREPARATION

[4] ✓ Os df.head()

***
Timestamp age Gender Purchase_Frequency Purchase_Categories Personalized_Recommendation_Frequency Browsing_Frequency Product_Search_Method Search_Result_E
0 2023/06/08 7:50:55 PM GMT+5:30 65 Prefer not to say Less than once a month Clothing and Fashion Yes Multiple times a day Keyword M
1 2023/06/09 9:37:44 AM GMT+5:30 20 Male Once a week Groceries and Gourmet Food;Beauty and Personal... No Rarely Filter M
2 2023/06/11 11:26:54 PM GMT+5:30 42 Male Once a week Groceries and Gourmet Food;Beauty and Personal... Sometimes Few times a week Keyword M
3 2023/06/08 5:17:10 PM GMT+5:30 65 Others Once a month Beauty and Personal Care;Clothing and Fashion;... No Few times a month Filter M
4 2023/06/11 10:59:30 PM GMT+5:30 45 Female Once a week Beauty and Personal Care;Clothing and Fashion;... Sometimes Few times a month NaN M

5 rows x 24 columns

```

```
df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 24 columns):
#   Column                                     Non-Null Count  Dtype
---  ---
0   Timestamp                                800 non-null    object
1   age                                       800 non-null    int64
2   Gender                                   800 non-null    object
3   Purchase_Frequency                      800 non-null    object
4   Purchase_Categories                     800 non-null    object
5   Personalized_Recommendation_Frequency  800 non-null    object
6   Browsing_Frequency                     800 non-null    object
7   Product_Search_Method                   643 non-null    object
8   Search_Result_Exploration               800 non-null    object
9   Customer_Reviews_Importance             800 non-null    int64
10  Add_to_Cart_Browsing                    800 non-null    object
11  Cart_Completion_Frequency               800 non-null    object
12  Cart_Abandonment_Factors                800 non-null    object
13  Saveforlater_Frequency                  800 non-null    object
14  Review_Left                             800 non-null    object
15  Review_Reliability                      800 non-null    object
16  Review_Helpfulness                      800 non-null    object
17  Personalized_Recommendation_Frequency  800 non-null    int64
18  Recommendation_Helpfulness              800 non-null    object
19  Rating_Accuracy                         800 non-null    int64
20  Shopping_Satisfaction                   800 non-null    int64
21  Service_Appreciation                    800 non-null    object
22  Improvement_Areas                       800 non-null    object
23  transaction                             800 non-null    int64

dtypes: int64(6), object(18)
memory usage: 150.1+ KB
```

[8]

✓ 0s



```
df.isnull().sum()
```



...

0

Timestamp

0

age

0

Gender

0

Purchase_Frequency

0

Purchase_Categories

0

Personalized_Recommendation_Frequency

0

Browsing_Frequency

0

Product_Search_Method

157

Search_Result_Exploration

0

Customer_Reviews_Importance

0

Add_to_Cart_Browsing

0

Cart_Completion_Frequency

0

Cart_Abandonment_Factors

0

Saveforlater_Frequency

0

Review_Left

0

Review_Reliability

0

Review_Helpfulness

0

Personalized_Recommendation_Frequency

0

Recommendation_Helpfulness

0

```
[9] df.columns = df.columns.str.strip()
    print(df.columns)

--- Index(['Timestamp', 'age', 'Gender', 'Purchase_Frequency',
        'Purchase_Categories', 'Personalized_Recommendation_Frequency',
        'Browsing_Frequency', 'Product_Search_Method',
        'Search_Result_Exploration', 'Customer_Reviews_Importance',
        'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',
        'Cart_Abandonment_Factors', 'Saveforlater_Frequency', 'Review_Left',
        'Review_Reliability', 'Review_Helpfulness',
        'Personalized_Recommendation_Frequency', 'Recommendation_Helpfulness',
        'Rating_Accuracy', 'Shopping_Satisfaction', 'Service_Appreciation',
        'Improvement_Areas', 'transaction'],
        dtype='object')

[10] df['Product_Search_Method'] = df['Product_Search_Method'].fillna('Keyword')

[11] df['Product_Search_Method'] = df['Product_Search_Method'].str.title()

[12] df['Customer_Reviews_Importance'] = df['Customer_Reviews_Importance'].astype(float)

[13] df['Rating_Accuracy'] = df['Rating_Accuracy'].astype(float)

[14] df['Shopping_Satisfaction'] = df['Shopping_Satisfaction'].astype(float)
```

df.dtypes

	0
Timestamp	object
age	int64
Gender	object
Purchase_Frequency	object
Purchase_Categories	object
Personalized_Recommendation_Frequency	object
Browsing_Frequency	object
Product_Search_Method	object
Search_Result_Exploration	object
Customer_Reviews_Importance	float64
Add_to_Cart_Browsing	object
Cart_Completion_Frequency	object
Cart_Abandonment_Factors	object
Saveforlater_Frequency	object
Review_Left	object
Review_Reliability	object
Review_Helpfulness	object
Personalized_Recommendation_Frequency	int64
Recommendation_Helpfulness	object

Task_2:-Descriptive_Behavior_Analysis

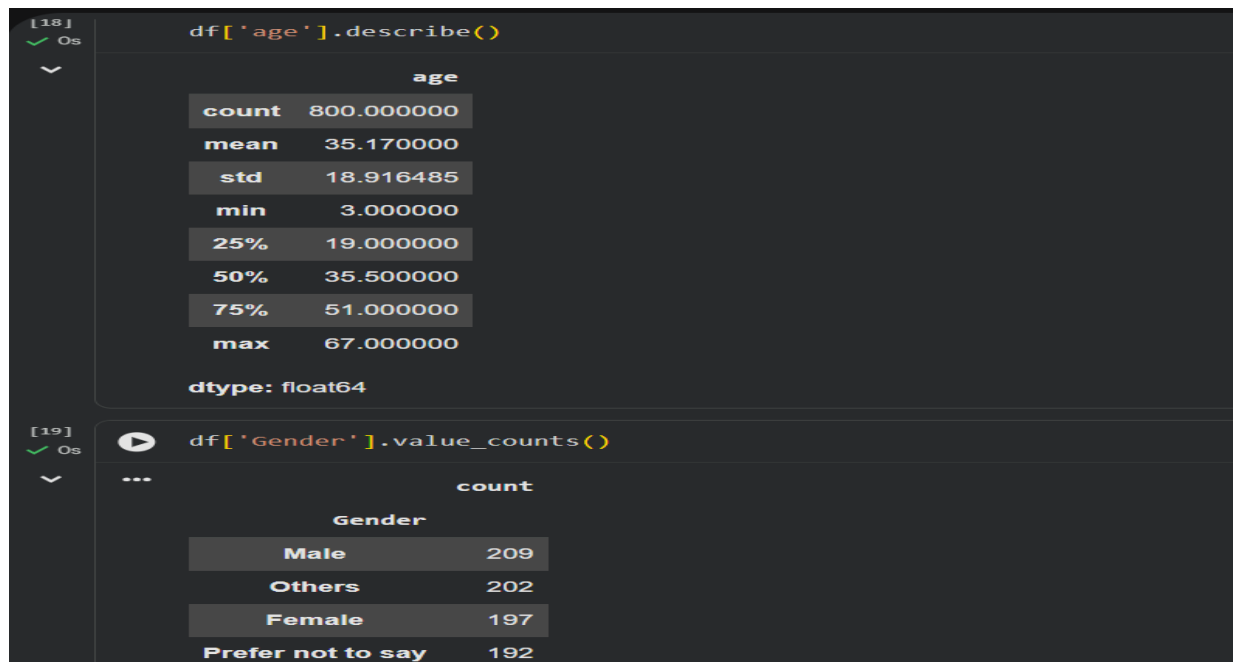
In this task, customer demographics and shopping behavior were analyzed using descriptive statistics and visualizations. Customer age distribution and gender participation were studied to understand the customer base. Purchase frequency analysis helped identify how often customers shop on Amazon. Popular product categories such as electronics and fashion were identified using bar charts and graphs. Also, Mean and median values were calculated for shopping satisfaction, recommendation helpfulness, and rating accuracy to understand the overall customer experience and satisfaction level.

```
TASK_2:-DESCRIPTIVE_BEHAVIOR_ANALYSIS
```

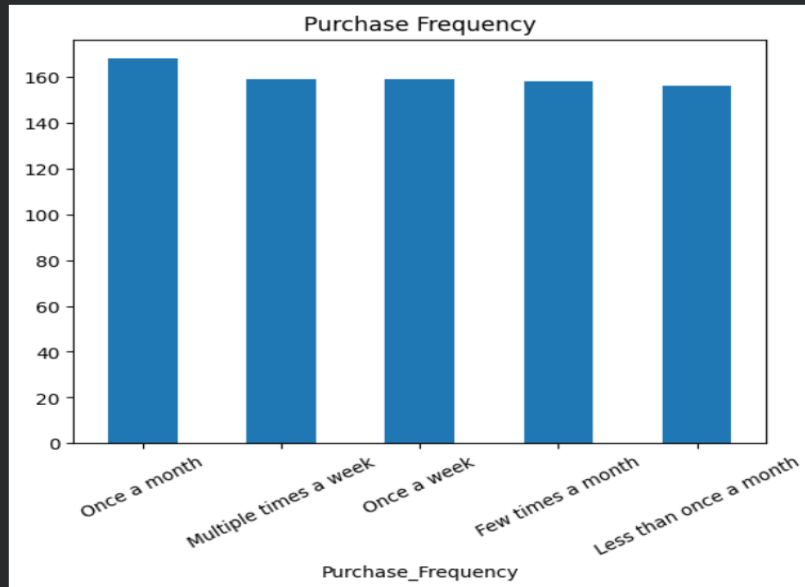
```
[17] ✓ Os df.head()
```

	Timestamp	age	Gender	Purchase_Frequency	Purchase_Categories	Personalized_Recommendation_Frequency	Browsing_Frequency	Product_Search_Method	Search_Result_E
0	2023/06/08 7:50:55 PM GMT+5:30	65	Prefer not to say	Less than once a month	Clothing and Fashion	Yes	Multiple times a day	Keyword	Mu
1	2023/06/09 9:37:44 AM GMT+5:30	20	Male	Once a week	Groceries and Gourmet Food;Beauty and Personal...	No	Rarely	Filter	Mu
2	2023/06/11 11:26:54 PM GMT+5:30	42	Male	Once a week	Groceries and Gourmet Food;Beauty and Personal...	Sometimes	Few times a week	Keyword	Mu
3	2023/06/08 5:17:10 PM GMT+5:30	65	Others	Once a month	Beauty and Personal Care;Clothing and Fashion;...	No	Few times a month	Filter	Mu
4	2023/06/11 10:59:30 PM GMT+5:30	45	Female	Once a week	Beauty and Personal Care;Clothing and Fashion;...	Sometimes	Few times a month	Keyword	Mu

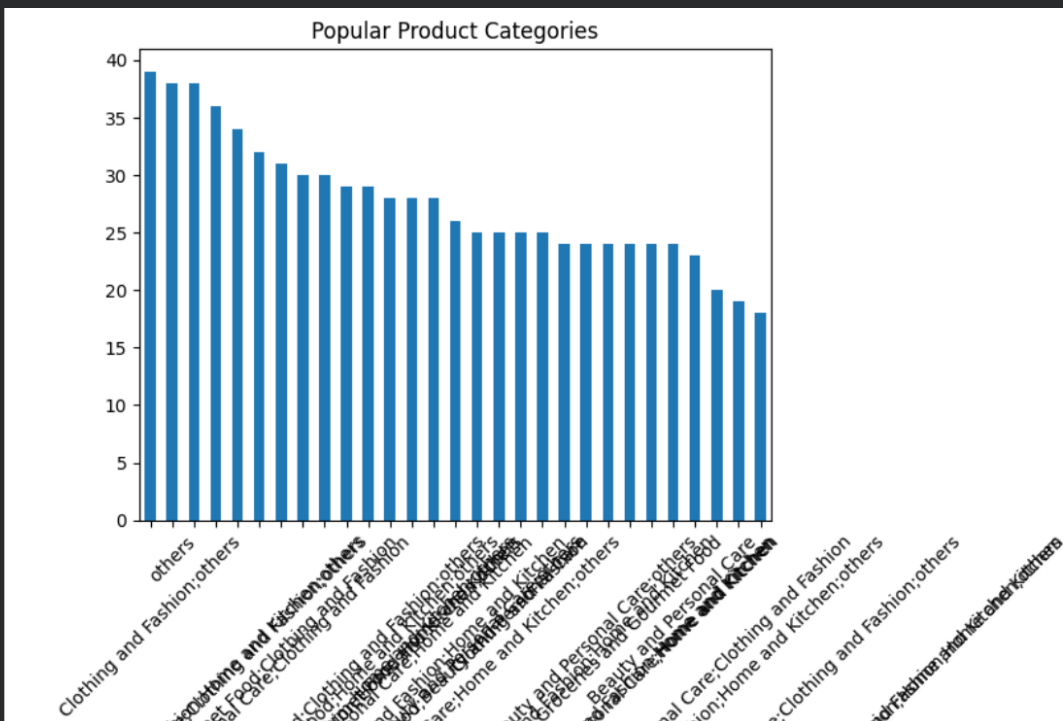
5 rows x 24 columns



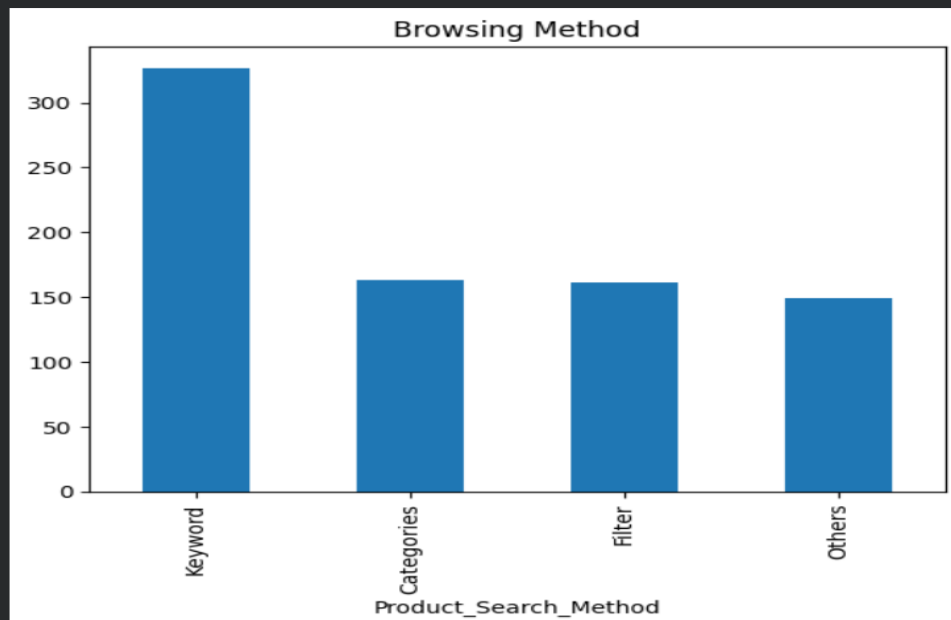
```
[22]
✓ Os
df['Purchase_Frequency'].value_counts().plot(kind='bar')
plt.title('Purchase Frequency')
plt.xticks(rotation=30)
plt.show()
```



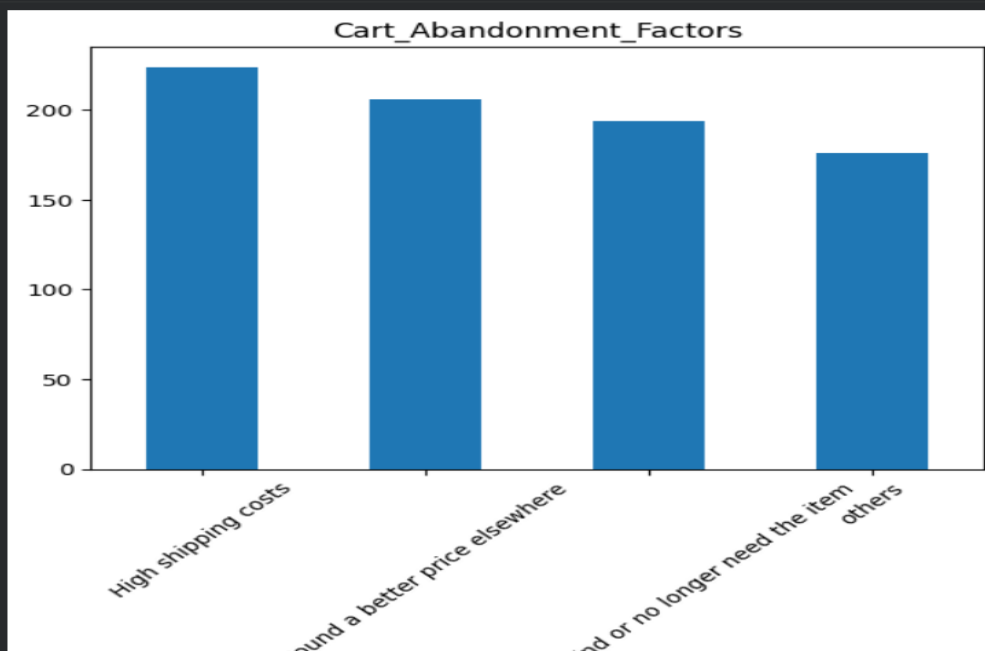
```
[24]
✓ Os
df['Purchase_Categories'].value_counts().plot(kind='bar')
plt.title('Popular Product Categories')
plt.xticks(rotation=45)
plt.show()
```



```
[26] df['Product_Search_Method'].value_counts().plot(kind='bar')  
✓ Os plt.title('Browsing Method')  
plt.show()
```



```
[28] df['Cart_Abandonment_Factors'].value_counts().plot(kind='bar')  
✓ Os plt.title('Cart_Abandonment_Factors')  
plt.xticks(rotation=45)  
plt.show()
```



Task_3:-Customer_Segmentation_and_Profilling

Task 3, focused on dividing customers into different groups based on their shopping behavior and satisfaction levels. Customers who frequently purchased products and showed high satisfaction levels were classified as Frequent Buyers. Customers who showed low satisfaction or high cart abandoned behavior were classified as At-Risk Customer. K-Means clustering was also used to group customers with similar behavioral patterns based on numerical variables such as satisfaction and rating accuracy.

```
TASK_3:-CUSTOMER_SEGAMENTATION_AND_PROFILLING

[39] ✓ 0s df['Purchase_Frequency'].unique()
      ▼
      array(['Less than once a month', 'Once a week', 'Once a month',
            'Few times a month', 'Multiple times a week'], dtype=object)

[40] ✓ 0s df['Shopping_Satisfaction'].unique()
      ▼
      array([4., 5., 3., 2., 1.])

[41] ✓ 0s ▶ conditions = [
            (df['Purchase_Frequency'] == 'Once a week') &
            (df['Shopping_Satisfaction'] >= 4),

            (df['Purchase_Frequency'] == 'Once a month') &
            (df['Shopping_Satisfaction'] >= 3),

            (df['Shopping_Satisfaction'] <= 2)
        ]

[42] ✓ 0s choices = [
            'Frequent Buyer',
            'Occasional Shopper',
            'At-Risk Customer'
        ]

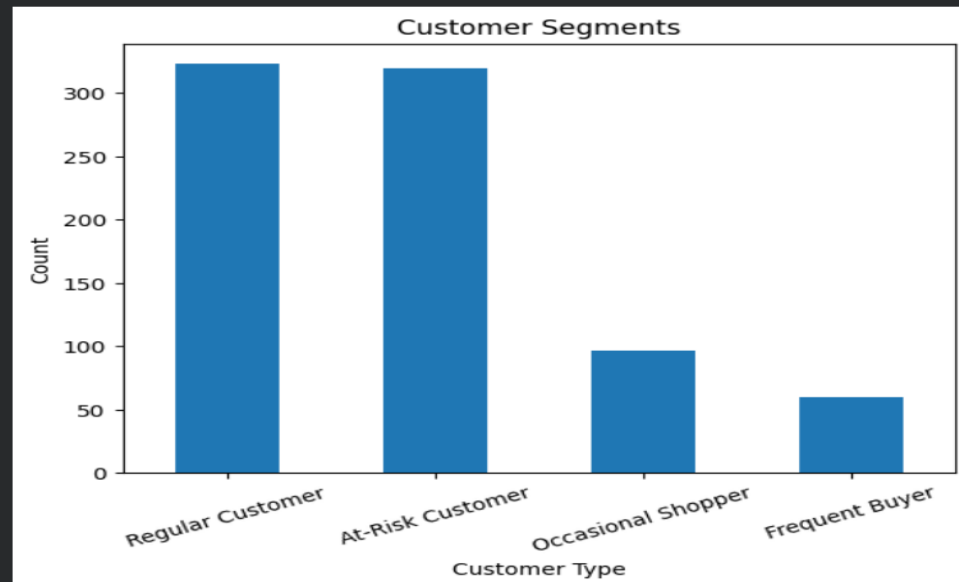
[43] ✓ 0s df['Customer_Profile'] = np.select(conditions, choices, default='Regular Customer')

[44] df['Customer_Profile'].value_counts()
```


[46]
✓ Os



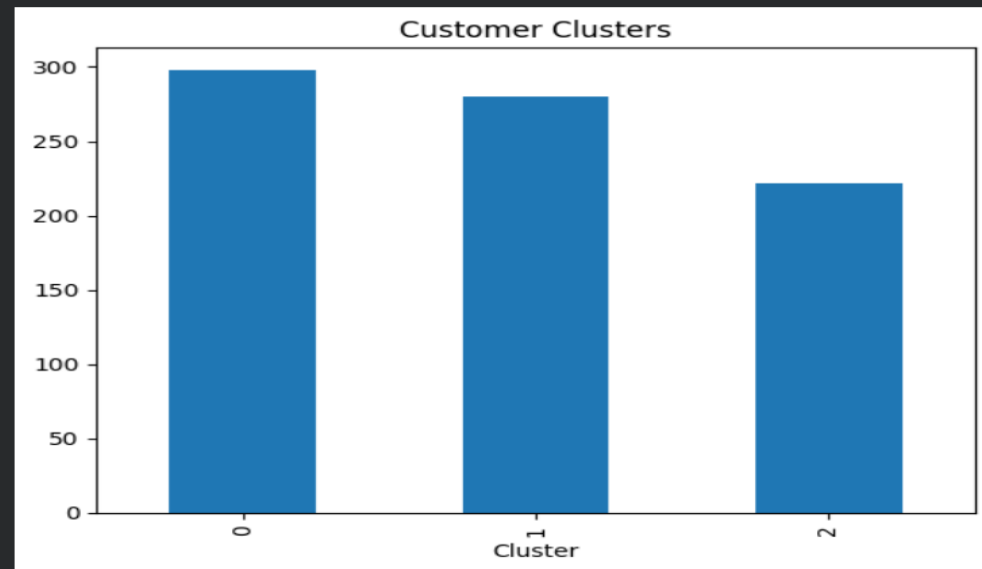
```
df['Customer_Profile'].value_counts().plot(kind='bar')  
plt.title('Customer Segments')  
plt.xlabel('Customer Type')  
plt.ylabel('Count')  
plt.xticks(rotation=20)  
plt.show()
```



[51]
✓ Os



```
df['cluster'].value_counts().plot(kind='bar')  
plt.title('Customer Clusters')  
plt.show()
```



[52]
✓ Os

```
df.to_csv('Amazon_Task_3_Customer_Segmentation.csv', index=False)
```

Task_4:-Recommendation_and_Review_Analysis

In this task, the relationship between recommendations, customer reviews, and satisfaction levels was analyzed. The analysis showed that customers who found recommendations helpful generally reported higher shopping satisfaction. Review reliability also had a positive effect on customer trust and rating accuracy. Based on these findings, several suggestions were identified to improve Amazon's recommendation system, such as providing more personalized recommendations and improving review authenticity.

Relationship Between Recommendation Helpfulness and Shopping Satisfaction:-

Recommendation helpfulness has a positive relationship with shopping satisfaction. Customers who found Amazon's recommendations useful and relevant were generally more satisfied with their shopping experience. Helpful recommendations made it easier for customers to find products according to their interests and needs, which improved their overall experience. This indicates that an effective recommendation system can increase customer engagement and improve customer satisfaction.

Actionable Insights to improve Amazon's Recommendation System:-

1. Amazon should provide more personalized recommendations based on customer interests and previous purchases.
2. Amazon can improve recommendation quality by understanding customer preferences and shopping behavior more effectively.

TASK_4:-RECOMMENDATION_AND_REVIEW_INSIGHTS

[53]
✓ 0s

```
df['Recommendation_Helpfulness'].unique()
```

```
array([1, 0, 2])
```

[54]
✓ 0s

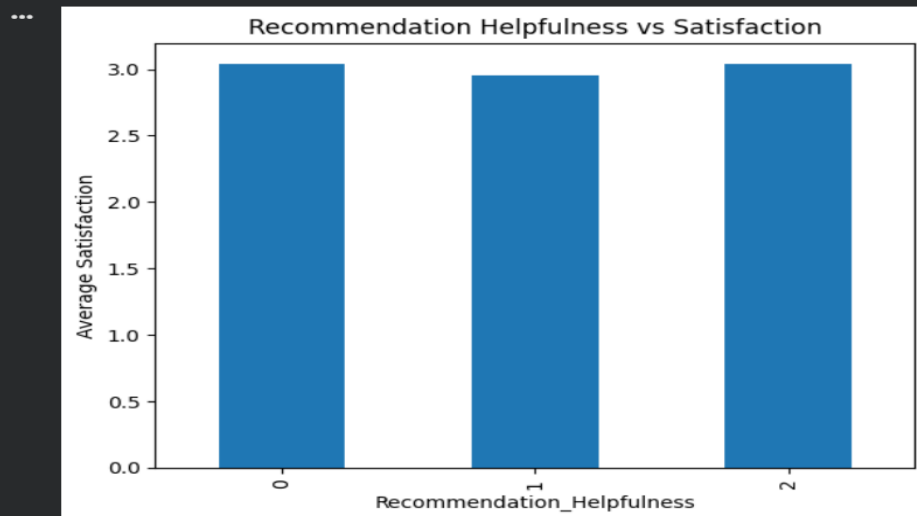
```
df.groupby('Recommendation_Helpfulness')['Shopping_Satisfaction'].mean()
```

Shopping_Satisfaction	
Recommendation_Helpfulness	
0	3.041985
1	2.955224
2	3.040741

dtype: float64

[55]
✓ 0s

```
df.groupby('Recommendation_Helpfulness')['Shopping_Satisfaction'].mean().plot(kind='bar')  
plt.title('Recommendation Helpfulness vs Satisfaction')  
plt.ylabel('Average Satisfaction')  
plt.show()
```



[56]
✓ 0s

```
df['Review_Reliability'].unique()
```

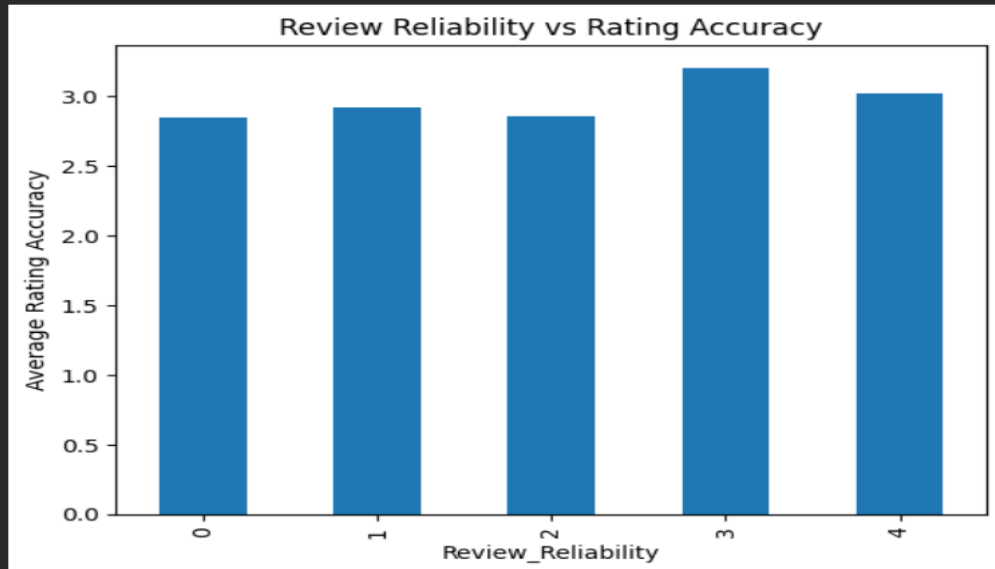
```
array(['Moderately', 'Heavily', 'Occasionally', 'Rarely', 'Never'],  
      dtype=object)
```

[60]
✓ Os

```
df.groupby('Review_Reliability')['Rating_Accuracy'].mean().plot(kind='bar')  
plt.title('Review Reliability vs Rating Accuracy')  
plt.ylabel('Average Rating Accuracy')  
plt.show()
```

▼

...

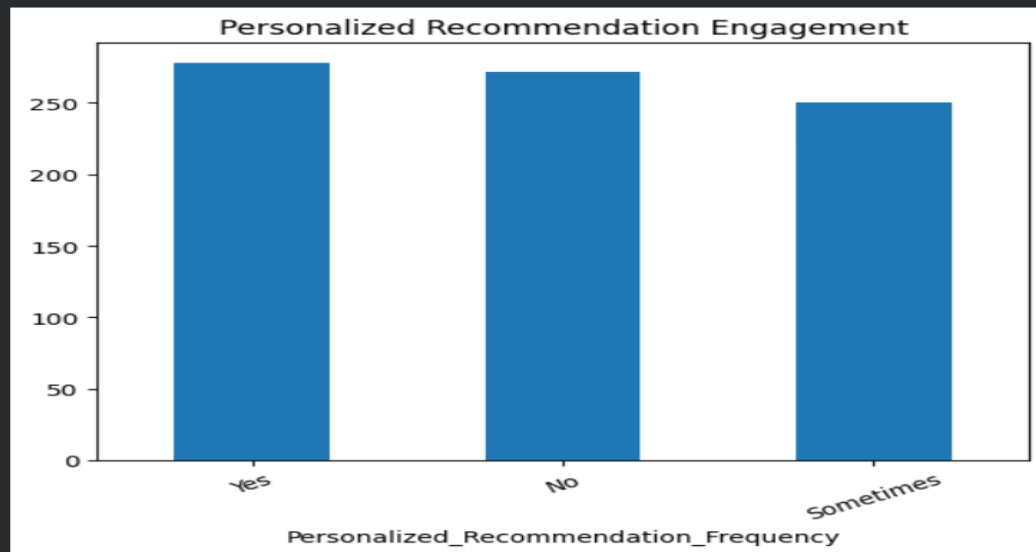


[65]
✓ Os

```
df['Personalized_Recommendation_Frequency'].value_counts().plot(kind='bar')  
plt.title('Personalized Recommendation Engagement')  
plt.xticks(rotation=25)  
plt.show()
```

▼

...

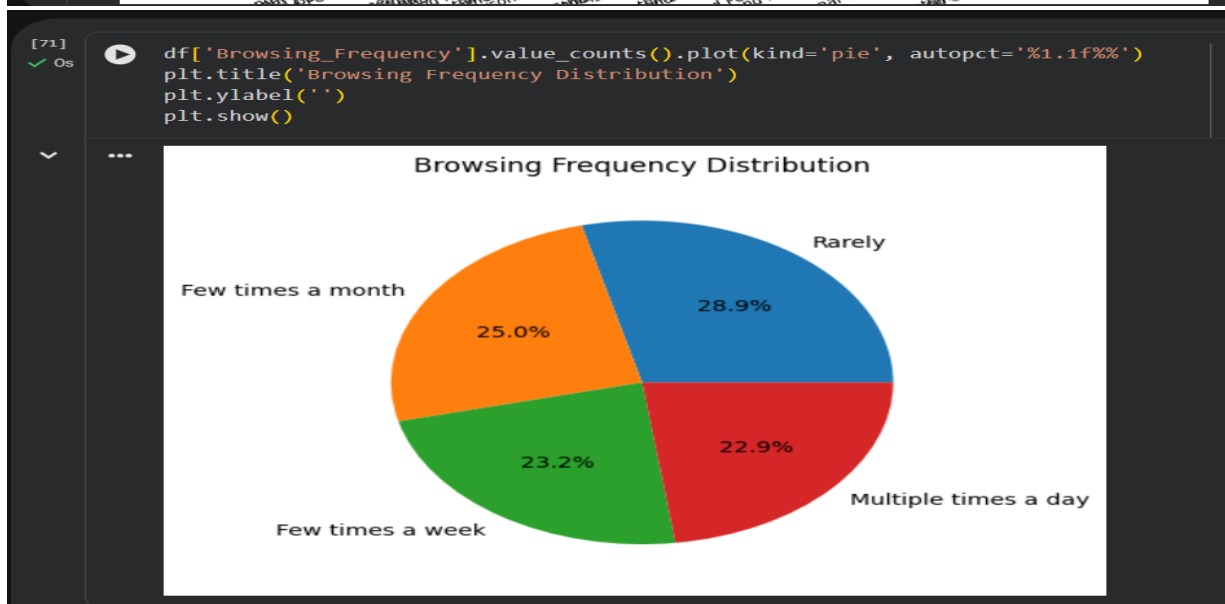
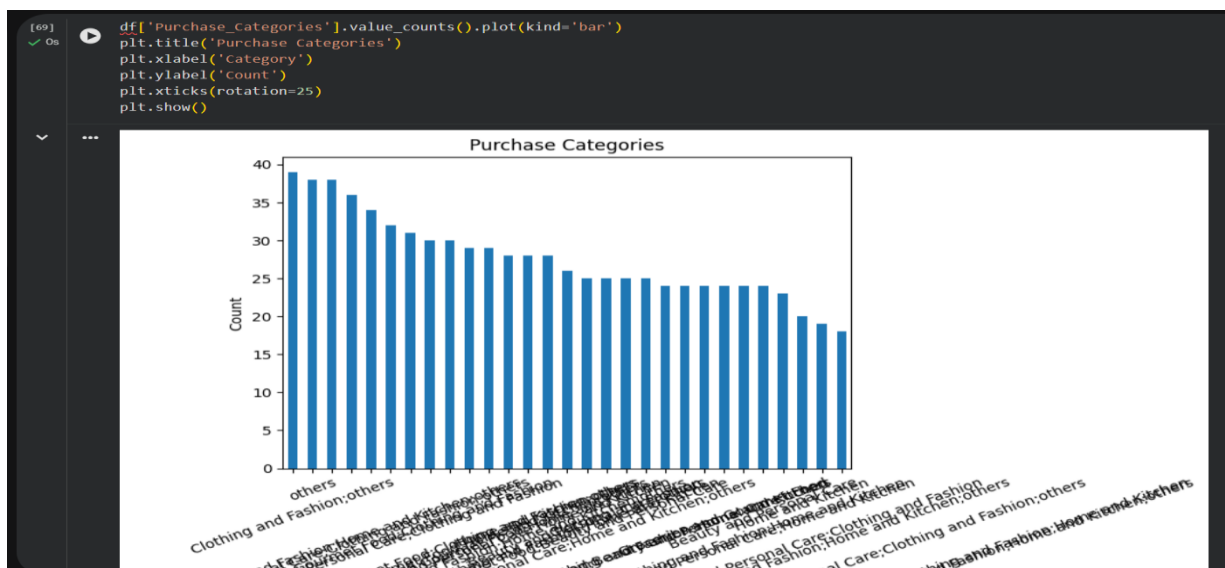


[66]
✓ Os

```
df.to_csv('Amazon_Task_4_Recommendation.csv', index=False)
```

Task_5:-Visualization_Reporting

Task 5, focused on presenting the analysis results through visualizations and reporting techniques. Different charts such as bar charts, pie charts, and heatmaps were created to visually represent customer behavior and shopping trends. Also, it used to Purchase categories, browsing frequency, customer satisfaction levels, and recommendation effectiveness were visualized for easier interpretation. Heatmaps were used to identify relationships between important numerical variables such as ratings, recommendations, and satisfaction levels.



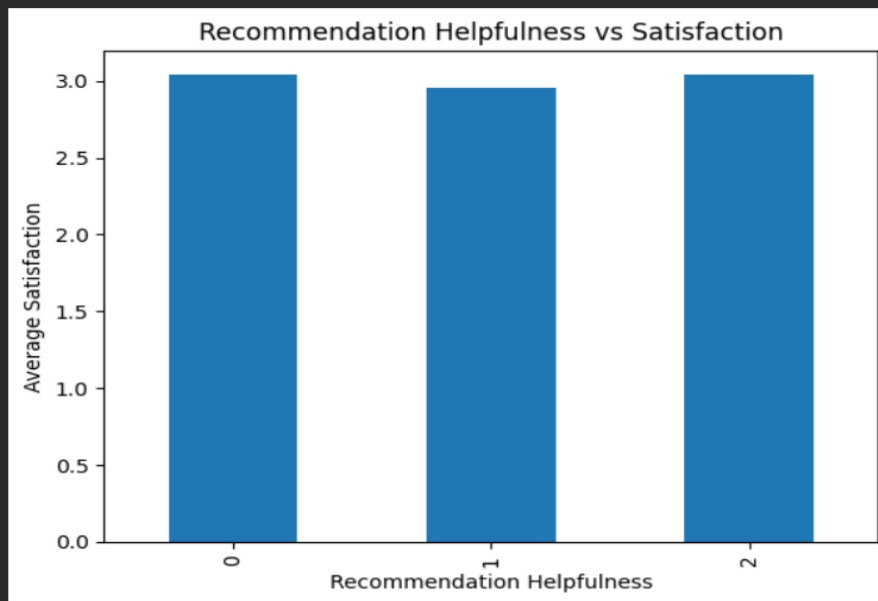
[72]
✓ 0s

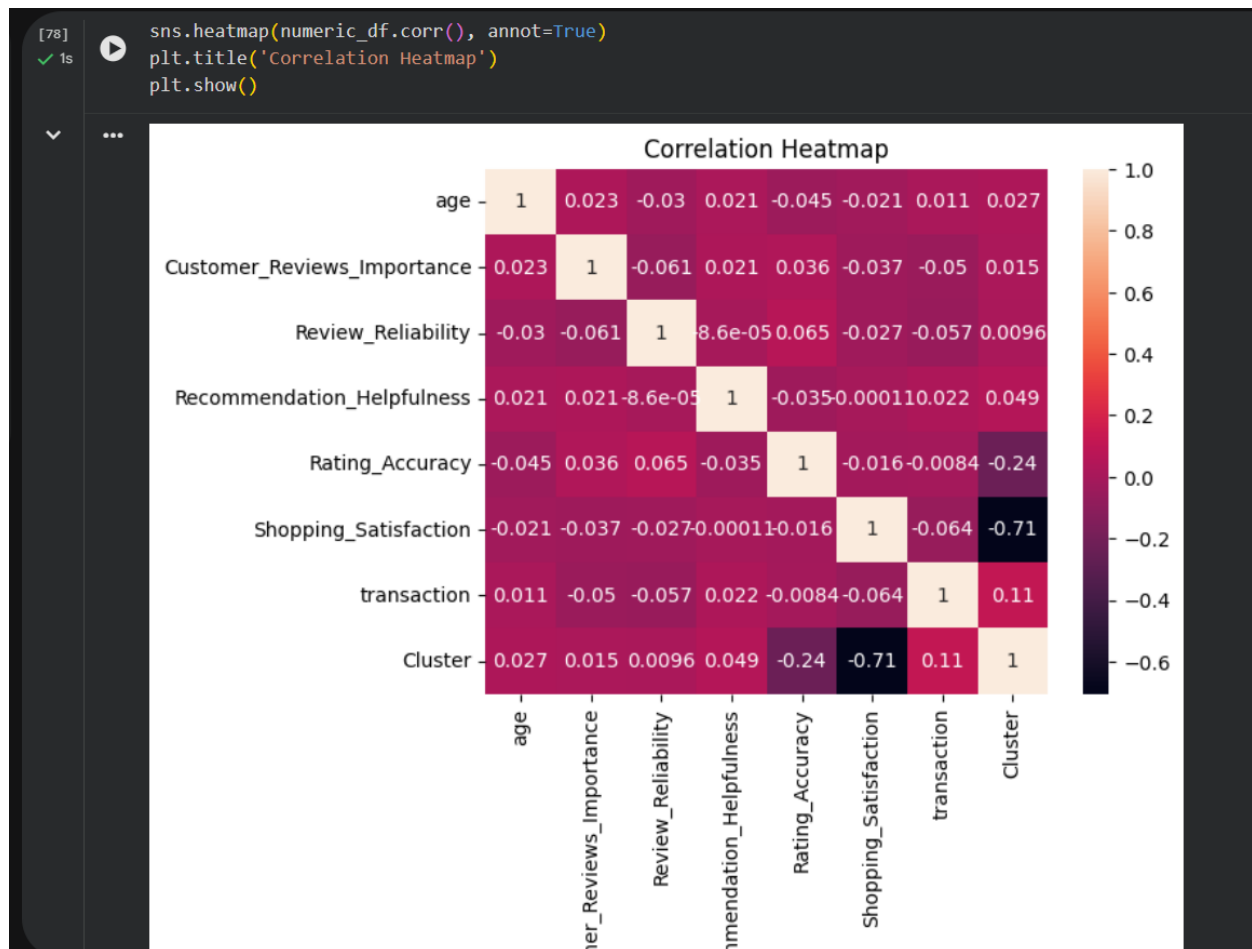
```
df['Shopping_Satisfaction'].value_counts().sort_index().plot(kind='bar')  
plt.title('Shopping Satisfaction Levels')  
plt.xlabel('Satisfaction Rating')  
plt.ylabel('Number of Customers')  
plt.show()
```



[75]
✓ 0s

```
df.groupby('Recommendation_Helpfulness')['Shopping_Satisfaction'].mean().plot(kind='bar')  
plt.title('Recommendation Helpfulness vs Satisfaction')  
plt.xlabel('Recommendation Helpfulness')  
plt.ylabel('Average Satisfaction')  
plt.show()
```





Conclusion:-

This concluded that the project successfully analyzed Amazon customer behavior using Python-based data analysis and visualization techniques. The project provided valuable insights into customer satisfaction, recommendation systems, shopping preferences, and behavioral patterns. The findings can help improve recommendation systems, customer engagement strategies, and overall shopping experience on e-commerce platforms

Project Drive Link:-

[https://drive.google.com/file/d/1B-sidl4bawvi1vglrDKb-9OeEfovU-VR/view?usp=drive link](https://drive.google.com/file/d/1B-sidl4bawvi1vglrDKb-9OeEfovU-VR/view?usp=drive_link)